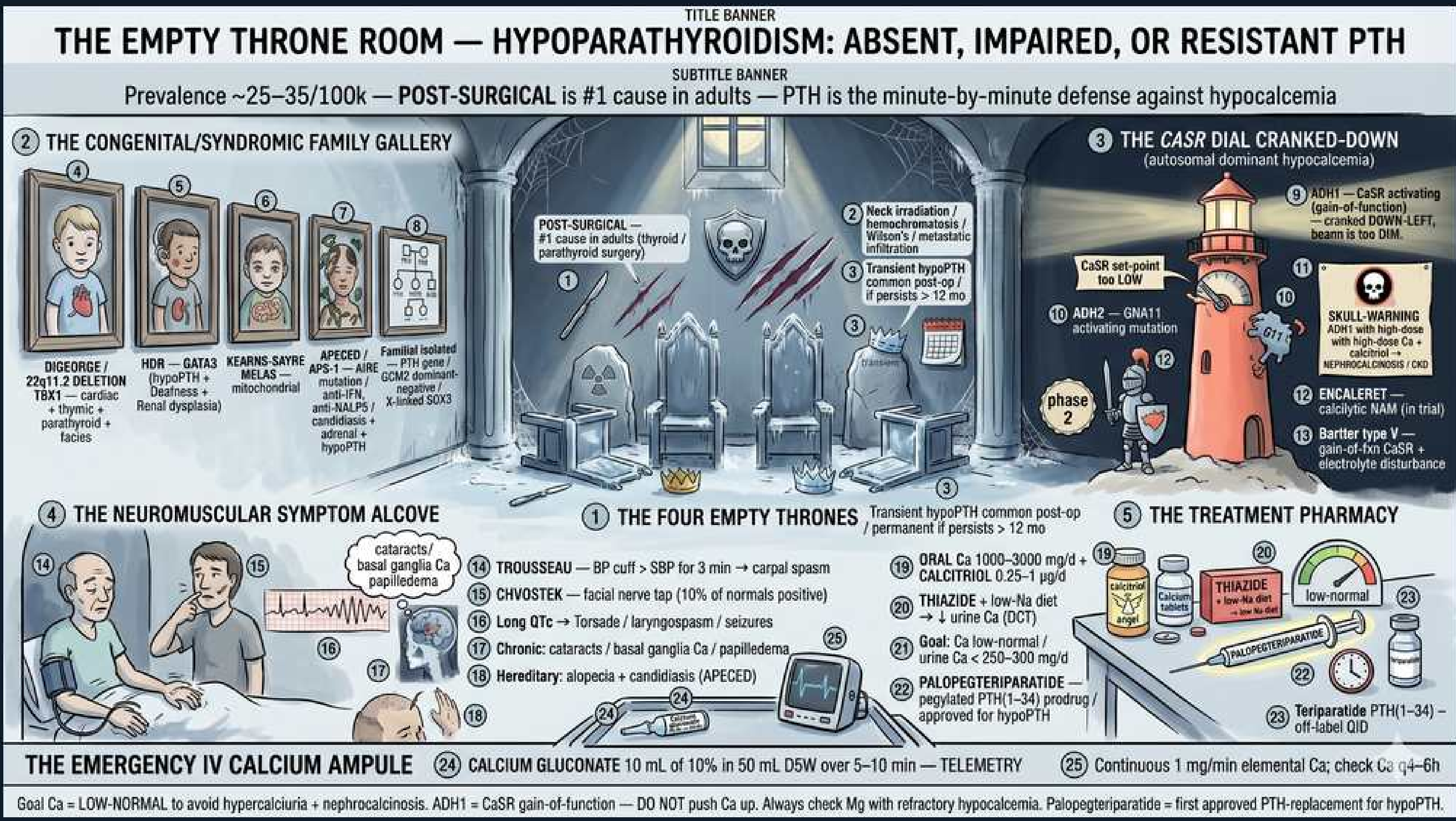
Hypoparathyroidism — Absent, Impaired, or Resistant PTH



1. Post-surgical — #1 in adults

WHAT YOU SEE

Severed parathyroid / scalpel

WHAT IT ENCODES

Post-surgical hypoparathyroidism (after thyroidectomy, parathyroidectomy, or neck dissection) is the #1 cause in adults. Transient hypoPTH is common post-op; permanent persists >6–12 months.

BOARD TRAP

Hungry-bone syndrome after parathyroidectomy ALSO causes post-op hypocalcemia but with NORMAL/high PTH and low phosphate — distinguish from true surgical hypoPTH (low PTH).

2. DiGeorge — 22q11.2 deletion

WHAT YOU SEE

Congenital family portrait, child

WHAT IT ENCODES

DiGeorge (22q11.2 del) = absent parathyroids + thymic aplasia (T-cell immunodeficiency) + conotruncal cardiac defects + facial anomalies. CATCH-22.

BOARD TRAP

Hypocalcemic seizures in a neonate with recurrent infections and a cardiac defect = DiGeorge, not isolated hypoPTH.

3. Autoimmune — APECED/APS-1

WHAT YOU SEE

APECED family figure

WHAT IT ENCODES

APS-1 (APECED, AIRE gene) triad: chronic mucocutaneous candidiasis + hypoPARAthyroidism + Addison adrenal insufficiency. Autoimmune destruction of parathyroids.

BOARD TRAP

Anti-NALP5 antibodies target parathyroid; new hypocalcemia + candidiasis + later Addison = APS-1, an autosomal recessive AIRE mutation.

4. Activating CaSR — ADH (GNA11/CaSR)

WHAT YOU SEE

CaSR dial cranked down

WHAT IT ENCODES

Autosomal dominant hypocalcemia (ADH): GAIN-of-function CaSR (or GNA11) mutation → the receptor senses a LOWER Ca set-point → suppresses PTH at normal/low Ca → hypocalcemia with INAPPROPRIATELY low PTH and often hypercalciuria.

BOARD TRAP

This is the MIRROR of FHH (loss-of-function CaSR). Aggressive Ca repletion in ADH worsens hypercalciuria/nephrocalcinosis/stones — treat cautiously, often with thiazide + low-normal Ca goal.

5. Skull X-ray / basal ganglia calcification

WHAT YOU SEE

Skull with calcified densities

WHAT IT ENCODES

Chronic hypoparathyroidism / high phosphate causes ectopic calcification: basal ganglia (Fahr syndrome), cataracts, and dental/skull changes. Long-standing hyperphosphatemia drives metastatic calcification.

BOARD TRAP

Posterior subcapsular cataracts and basal-ganglia calcification on CT in a young patient point to chronic hypoPTH — not a primary neurologic disorder.

6. Trousseau sign

WHAT YOU SEE

BP cuff → carpal spasm

WHAT IT ENCODES

Trousseau sign: inflating a BP cuff above systolic for ~3 min provokes carpopedal spasm — a sign of neuromuscular irritability from hypocalcemia.

BOARD TRAP

Trousseau is MORE sensitive/specific than Chvostek; Chvostek (facial twitch) is present in ~10% of normals, so a positive Chvostek alone is weaker evidence.

7. Chvostek sign

WHAT YOU SEE

Facial-nerve tap, cheek twitch

WHAT IT ENCODES

Chvostek sign: tapping over the facial nerve causes ipsilateral facial muscle twitch in hypocalcemia. Less specific than Trousseau.

BOARD TRAP

~10% of normocalcemic people have a positive Chvostek — do not diagnose hypocalcemia on Chvostek alone; confirm with ionized/corrected Ca.

8. Long QT — telemetry

WHAT YOU SEE

ECG strip with long QT

WHAT IT ENCODES

Hypocalcemia prolongs the QT interval (ST/QT lengthening) → risk of torsades and arrhythmia. Get an ECG and telemetry in severe hypocalcemia.

BOARD TRAP

Hypocalcemia prolongs QT; hyperCALCEMIA SHORTENS QT — a frequently swapped board fact.

9. Labs: low Ca, high PO4, low PTH

WHAT YOU SEE

Four empty thrones (low PTH)

WHAT IT ENCODES

Classic hypoPTH biochemistry: LOW calcium, HIGH phosphate, and LOW/inappropriately-normal PTH, with low 1,25-D (PTH normally drives renal 1-alpha-hydroxylase). Check magnesium.

BOARD TRAP

High phosphate WITH low PTH = hypoPTH; in CKD you also get low Ca/high PO4 but PTH is HIGH (secondary hyperPTH) — phosphate-PTH pairing separates them.

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10. Check magnesium — refractory hypoCa

WHAT YOU SEE

Mg flask warning

WHAT IT ENCODES

Always check Mg in hypocalcemia. Severe HYPOmagnesemia (PPIs, alcohol, diarrhea) causes functional hypoPTH — it impairs PTH secretion AND causes end-organ PTH resistance.

BOARD TRAP

Refractory hypocalcemia that won't correct with Ca alone = check/replace MAGNESIUM first; giving Ca without fixing Mg fails.

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11. Calcium gluconate — acute IV

WHAT YOU SEE

IV calcium ampule

WHAT IT ENCODES

Acute symptomatic hypocalcemia: IV calcium GLUCONATE (1–2 g of 10% over 10–20 min in D5W), then a calcium infusion. Gluconate preferred peripherally (less caustic than chloride).

BOARD TRAP

Calcium CHLORIDE has 3x the elemental Ca but is vesicant — reserve for central line/codes; gluconate is the peripheral choice.

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12. Calcitriol + oral Ca

WHAT YOU SEE

Pharmacy: calcitriol + thiazide

WHAT IT ENCODES

Chronic hypoPTH therapy: oral calcium + ACTIVE vitamin D (calcitriol), because absent PTH cannot activate renal 1-alpha-hydroxylase. A thiazide reduces urinary Ca losses and hypercalciuria.

BOARD TRAP

You must use calcitriol (active), NOT plain cholecalciferol alone, in hypoPTH — the kidney cannot 1-alpha-hydroxylate without PTH.

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13. Goal Ca low-normal

WHAT YOU SEE

Target set-point banner

WHAT IT ENCODES

Treatment goal is LOW-normal serum Ca to avoid symptoms while minimizing hypercalciuria, nephrocalcinosis, and stones (no PTH to reabsorb urinary Ca).

BOARD TRAP

Aiming for mid/high-normal Ca in hypoPTH drives hypercalciuria and kidney stones — accept low-normal Ca on purpose.

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14. Teriparatide / PTH(1–34)

WHAT YOU SEE

Teriparatide syringe

WHAT IT ENCODES

Recombinant PTH (teriparatide / PTH 1–34, or palopegteriparatide as a long-acting PTH replacement) can be used in refractory hypoPTH to restore the missing hormone and reduce Ca/calcitriol burden.

BOARD TRAP

This is hormone REPLACEMENT in hypoPTH — the opposite indication from osteoporosis where teriparatide is anabolic; it directly fixes the deficient hormone here.
